# Non-Graded Project: Cyber Deterrence and Resilience Mini-Challenge

# Learning Objectives

By the end of this project, learners will be able to:

- Analyze a real-world cyber threat challenge and identify its impact on organizational continuity.
- Evaluate potential defense and resilience solutions based on feasibility, benefits, and limitations within an enterprise network.
- Design a strategic cyber proposal that includes deterrence measures, technical requirements, and ethical/privacy safeguards.



# Description: Cyber Deterrence and Resilience Mini-Challenge

Cyberattacks are one of the leading causes of operational disruption among modern enterprises, often halting business functions and compromising sensitive data. In this project, you'll step into the role of a Chief Information Security Officer (CISO) tasked with solving a pressing vulnerability issue at Global-Logix, a multinational logistics company.

You'll explore how deterrence strategies (like deception) and resilience frameworks (like NIST) can harden the network, reduce the impact of a breach, and improve recovery efficiency without compromising user privacy or network speed. Using a structured approach, you'll identify the core problem, explore defense options, evaluate potential impacts, and address ethical considerations regarding data monitoring. Your final deliverable will be a concise mini-proposal outlining a deterrence and resilience solution tailored for real-world applications.

This challenge offers a hands-on opportunity to bridge technical knowledge with real organizational needs, helping you think critically about both strategic defense and responsibility in cybersecurity deployment.

## Structure of the Project:

**Title:** Cyber Deterrence and Resilience Mini-Challenge

**Tools:** ChatGPT, Claude, NIST CSF Guidelines

# Scenario

Sarah Chen, the VP of Operations at Global-Logix, faces increasing frustration with the overwhelming threat of ransomware and data theft. With a hybrid-cloud network and 500 remote endpoints, the IT team finds themselves constantly reacting to minor breaches, leading to burnout and reduced operational efficiency. The executive board is actively exploring strategic solutions to deter attackers and ensure rapid recovery while maintaining compliance. As an incoming CISO, you have been brought in to propose an innovative, resilient solution.

# Objective

Your goal is to design a cyber deterrence and resilience solution that alleviates the constant threat of operational downtime. This exercise will help you analyze defense applications, assess their impact, and consider ethical implications in enterprise monitoring.



# Follow these steps to complete this project

# Step 1: Identify Cybersecurity Challenge

Problem Statement: Sarah Chen and the IT team spend too much time reacting to successful phishing and malware intrusions rather than focusing on proactive continuity, leading to stress and operational delays. This challenge requires a strategic solution that deters attackers and streamlines recovery without compromising privacy or compliance.

## Write a one-paragraph problem statement describing:

- The impact of constant cyber threats on IT professionals.
- How this challenge affects business continuity and enterprise efficiency.
- Why an integrated deterrence and resilience solution is necessary to address this issue.

# Step 2: Analyze Defense/Resilience Options

- Explore potential strategic solutions that could address Sarah's security challenge.
  - Research potential applications (e.g., Honeypots, Automated Backups, ISO 22301) for your chosen challenge using course materials.
  - Note 2-3 possible strategic solutions.
  - List key features and requirements.

## Step 3: Evaluate Impact

Assess how the proposed solution would transform Global-Logix's workflow and operations.

## Workflow Impact

- Describe how the solution would change current reactive processes to proactive ones.
- Identify key stakeholders affected.

## Expected Benefits

- Estimate potential time/cost savings (e.g., improving Mean Time to Recovery).
- List 2-3 security quality improvements.

## Step 4: Consider Implementation Requirements

- Outline the technical and operational needs for successfully implementing the resilience solution at Global-Logix.
- List technical requirements (e.g., network segmentation, backup servers).
- Identify necessary staff training.
- Note potential integration points with existing systems.

## Step 5: Address Ethical Considerations

- Identify a key ethical concern and propose a safeguard to ensure employee privacy and security during threat monitoring.
  - Identify one key ethical concern (e.g., accidentally monitoring legitimate employee traffic with deception tools).
  - Propose one specific safeguard.
  - Consider data privacy implications.

## Organize information following this structure for peer review:

- **Title Page:** Project title, your name, date
- **Challenge Description:** Brief problem statement
- **Proposed Solution:** Deterrence/Resilience application description
- **Implementation Overview:** Key requirements
- **Ethical Considerations:** Main concern and safeguard
- **Recommendation:** One concrete next step

# Instructions for Documenting and Sharing the Project for Peer Review

## Document the Project

- Create your proposal using word processing software.
- Keep sections concise and focused.
- Save as PDF.

# Share the Project

- Upload to the "Peer Review" area
- Include a brief description of your chosen cybersecurity challenge

## Peer Review

- Review other students' proposals
- Provide constructive feedback on feasibility and ethical consideration

By completing this mini-challenge, you'll gain practical experience in applying cyber deterrence and resilience strategies to enterprise environments while considering both technical and ethical implications.